

David B. Hamilton
Vice President

440-280-5382

June 27, 2017
L-17-20710CFR50.73(a)(2)(v)(A)
10CFR50.73(a)(2)(v)(D)ATTN: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, DC 20555-0001SUBJECT:
Perry Nuclear Power Plant
Docket No. 50-440, License No. NPF-58
Licensee Event Report Submittal

Enclosed is Licensee Event Report (LER) 2017-002, "Loss of Safety Function Due to Main Turbine Bypass Valve Opening". There are no regulatory commitments contained in this submittal.

If there are any questions or if additional information is required, please contact Mr. Nicola Conicella, Manager – Regulatory Compliance, at (440) 280-5415.

Sincerely,

David B. Hamilton
Vice PresidentEnclosure:
LER 2017-002cc: NRC Project Manager
NRC Resident Inspector
NRC Region III

Enclosure
L-17-207

LER 2017-002

**LICENSEE EVENT REPORT (LER)**
(See Page 2 for required number of digits/characters for each block)(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME

Perry Nuclear Power Plant

2. DOCKET NUMBER

05000-440

3. PAGE

1 OF 4

4. TITLE:

Loss of Safety Function Due to Main Turbine Bypass Valve Opening

5. EVENT DATE

MONTH	DAY	YEAR
04	30	2017

6. LER NUMBER

YEAR	SEQUENTIAL NUMBER	REV NO.
2017	002	00

7. REPORT DATE

MONTH	DAY	YEAR
06	27	2017

8. OTHER FACILITIES INVOLVED

FACILITY NAME	DOCKET NUMBER
	05000
FACILITY NAME	DOCKET NUMBER
	05000

9. OPERATING MODE

1

11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)

☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☒ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ OTHER	Specify in Abstract below or in NRC Form 366A

12. LICENSEE CONTACT FOR THIS LER**LICENSEE CONTACT:**

Tony Kledzik – Regulatory Compliance

TELEPHONE NUMBER (Include Area Code)

440-280-6188

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANUFACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANUFACTURER	REPORTABLE TO EPIX
E	JL	SC	XX	Y					

14. SUPPLEMENTAL REPORT EXPECTED☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO**15. EXPECTED SUBMISSION DATE**

MONTH	DAY	YEAR

ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

On April 30, 2017, at 1818 hours, while the plant was at 100 percent rated thermal power, main turbine steam bypass valve number 1 partially opened. Power was subsequently lowered in an attempt to close the bypass valve. While lowering power the bypass valve would shut and then reopen and power would again be lowered. When power was lowered to approximately 74 percent the bypass valve remained closed. During the transient the reactor protection system (RPS) trip functions for the main turbine stop valve closure and turbine control valve fast closure scram were declared inoperable due to the opening of the bypass valve, which changes the bypass setpoint for those RPS trips. With the loss of RPS trip capability, a loss of safety function existed intermittently for approximately 37 minutes. The manual reactor trip function and other RPS functions remained operable.

Both channels of the rod withdrawal limiter (RWL) and the end of cycle reactor recirculation pump trip (EOC-RPT) function were also declared inoperable. These functions are credited in accident analysis and this also resulted in a loss of safety function in accordance with the plants Technical Specification bases.

The direct cause of the bypass valve opening was degradation of the Primary Low Value Gate (PLVG) card in the main turbine speed control circuit.

The safety significance of this event is considered to be small. This event is not considered a safety system functional failure as the specific functions were maintained and never bypassed during the event. This event is being reported under 50.73(a)(2)(v)(A) and 50.73(a)(2)(v)(D) for a loss of safety function.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Perry Nuclear Power Plant Unit 1	05000-440	2017	- 002	- 00

NARRATIVE

Energy Industry Identification System (EIIIS) codes are identified in the text as [XX].

INTRODUCTION

On April 30, 2017, at 1818 hours, while the plant was at 100 percent rated thermal power, main turbine steam bypass [JI] valve number 1 partially opened. Power was subsequently lowered in an attempt to close the bypass valve. While lowering power the bypass valve would shut and then reopen and power would again be lowered. When power was lowered to approximately 74 percent the bypass valve remained closed. During the transient the reactor protection system (RPS) [JC] trip functions for the main turbine stop valve closure and turbine control valve fast closure scram were declared inoperable due to the opening of the bypass valve which changes the bypass setpoint for those RPS trips. With the loss of RPS trip capability for this function a loss of safety function existed intermittently for approximately 37 minutes. The manual reactor trip function and other RPS functions remained operable.

Both channels of the rod withdrawal [AA] limiter (RWL) and the end of cycle reactor recirculation pump [AD] trip (EOC-RPT) function were also declared inoperable in accordance with the plants Technical Specification (TS) bases. These functions are credited in accident analysis, this also resulted in a loss of safety function.

On April 30, 2017, at 1910 hours, the action statements for TS 3.3.1.1 "Reactor Protection System (RPS) Instrumentation", 3.3.2.1 "Control Rod Block Instrumentation", and 3.3.4.1 "End of Cycle Recirculation Pump Trip (EOC-RPT) Instrumentation" were exited following confirmation of bypass valve closure. On May 3, 2017, at 1259 hours, event notification 52727 was made to the NRC Operations Center in accordance with 10CFR50.72(b)(3)(v)(A) and 10CFR50.72(b)(3)(v)(D) for the loss of safety function.

EVENT DESCRIPTION

On April 30, 2017, at 1818 hours, while the plant was at 100 percent rated thermal power main turbine steam bypass valve number 1 partially opened. Power was subsequently lowered in an attempt to close the bypass valve. While lowering power the bypass valve would shut and then reopen and power would again be lowered. When power was lowered to approximately 74 percent the bypass valve remained closed.

Certain TS functions are automatically bypassed based upon certain sensed reactor power levels. Four of those functions rely on the pressure in the first stage of the turbine to correlate reactor power. With a bypass valve open, the pressure in the first stage of the turbine may not be representative of true reactor power so these functions are declared inoperable. The following functions use the first stage pressure:

Turbine Stop Valve (TSV) Closure and Turbine Control Valve (TCV) Fast Closure Scram – TS 3.3.1.1:

Closure of the TSVs and TCVs results in the loss of a heat sink and produces reactor pressure, neutron flux, and heat flux transients that must be limited. Therefore, a reactor scram is initiated at the start of TSV and TCV closure in anticipation of the transients that would result from the closure of these valves. This function must be enabled at thermal power greater than or equal to 38 percent. This is normally accomplished automatically by pressure switches sensing turbine first stage pressure; therefore, to consider this function operable, the turbine bypass valves must remain shut at power greater than or equal to 38 percent.

End of Cycle Recirculation Pump Trip (EOC-RPT) – TS 3.3.4.1:

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NARRATIVE

To mitigate pressurization transient effects of a main turbine trip the EOC-RPT must trip the recirculation pumps from fast speed operation after initiation of initial closure movement of either the TSVs or the TCVs. The combined effects of this trip and a scram reduce fuel bundle power more rapidly than does a scram alone, resulting in an increased margin to the Minimum Critical Power Ratio (MCPR) safety limit. The EOC-RPT function is automatically disabled when turbine first stage pressure is less than 38 percent power. To consider this function operable, the turbine bypass valves must remain shut at power greater than or equal to 38 percent.

Rod Withdrawal Limiter (RWL) – TS 3.3.2.1:

The RWL is designed to prevent violation of the MCPR safety limit and the cladding 1 percent plastic strain fuel design limit that may result from a single control rod withdrawal error (RWE) event. The RWL is assumed to mitigate the consequences of an RWE event when operating greater than 33.3 percent power.

With any bypass valve open both channels of the listed functions are inoperable resulting in a loss of safety function. During the transient the following TS action statements were entered: TS 3.3.1.1 conditions A, B and C, 3.3.2.1 condition A and 3.3.4.1 conditions A and B. The required action for TS 3.3.1.1 C.1 requires restoration of RPS trip capability within 1 hour. TS 3.3.2.1 Action A.1 requires immediate suspension of control rod withdrawal and verification of no rod motion by 2 licensed operators. 3.3.4.1 Action B.1 requires restoration of EOC-RPT capability within 2 hours.

On April 30, 2017, at 1910 hours, the action statements for TS 3.3.1.1, 3.3.2.1, and 3.3.4.1 were exited following confirmation of bypass valve closure. On May 3, 2017, at 1259 hours event notification 52727 was made to the NRC Operations Center in accordance with 10CFR50.72(b)(3)(v)(A) and 10CFR50.72(b)(3)(v)(D) for the loss of safety function.

CAUSE

The direct cause of the bypass valve opening was degradation of the Primary Low Value Gate (PLVG) card in the main turbine speed control circuit [SC].

The problem solving team assembled for this issue evaluated monitoring equipment attached to the speed control portion of the Electro-Hydraulic system in order to determine the source of the inadvertent opening of the bypass valve. The failure mode was narrowed down to the turbine speed control portion of the circuitry based upon the alarms received during the transient. The decision was made to replace the Primary Low Value Gate and the Primary Frequency to Voltage Converter Speed Control cards. After the replacement was completed, the turbine was re-synchronized to the electrical grid. The installed monitoring equipment showed no evidence of an issue within the Electro-Hydraulic control systems and there were no further openings of the bypass valves.

The analysis of the removed cards is not yet completed, and if a change to the primary cause is warranted a LER revision will be submitted. Preliminary testing found a degradation of the PLVG card. During bench checks of the card it was discovered that an intermittent varying output signal was created with no changes to the input of the card. This intermittent change in output of the card would have caused the opening of the bypass valve. The second card that was replaced was the Primary Frequency to Voltage Converter card although initial analysis has identified no issues with this card.

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NARRATIVE**EVENT ANALYSIS**

The event is of small safety significance from the Probabilistic Risk Assessment (PRA) perspective as no PRA related mitigative components or strategies were affected by this event. While the corresponding functions were declared inoperable from a Technical Specification perspective, functionality was maintained and the plant was correspondingly controlled via proper actions to reduce power. The event did not result in an automatic trip.

With no transient and no impact to any PRA related components or functions, there would be no corresponding change (delta) in core damage frequency (CDF), and no corresponding change (delta) in the large early release frequency (LERF). The delta CDF and delta LERF values would therefore be well below the acceptable thresholds of $1.0E-06/\text{yr}$ and $1.0E-07/\text{yr}$, respectively, as discussed in Regulatory Guide 1.174. The risk of this event is therefore considered small in accordance with the Regulatory Guidance.

During the transient, the sensed reactor pressure at the turbine first stage did not drop low enough to impede the ability for the listed parameters to perform their function, therefore, this is not considered a safety system functional failure.

CORRECTIVE ACTIONS

The PLVG card was replaced following discovery of the failure. No other issues have been noted since replacement.

In addition to the card replacement, the mitigation strategy for the control system will be reviewed.

PREVIOUS SIMILAR EVENTS

A review of LERs and the corrective action database for the past three years identified no similar events.

COMMITMENTS

There are no regulatory commitments contained in this report. Actions described in this document represent intended or planned actions, are described for the NRC's information, and are not regulatory commitments.